VibroVision

From machine video to motion measurements, inspectable evidence and a human decision.

OpenCV AI Competition 2026, powered by AWS | 26 October 2026

Live application: d30icy4clt3p0c.cloudfront.net
Repository: github.com/emezac/vibrovision

Problem, users and approach

Maintenance engineers, researchers and educators can use ordinary video to inspect visible movement without placing a sensor on every region. VibroVision measures displacement with OpenCV 5, checks the measurement and uses a rule-based inspection agent to choose a further check, generate evidence or request review. It does not actuate machinery.

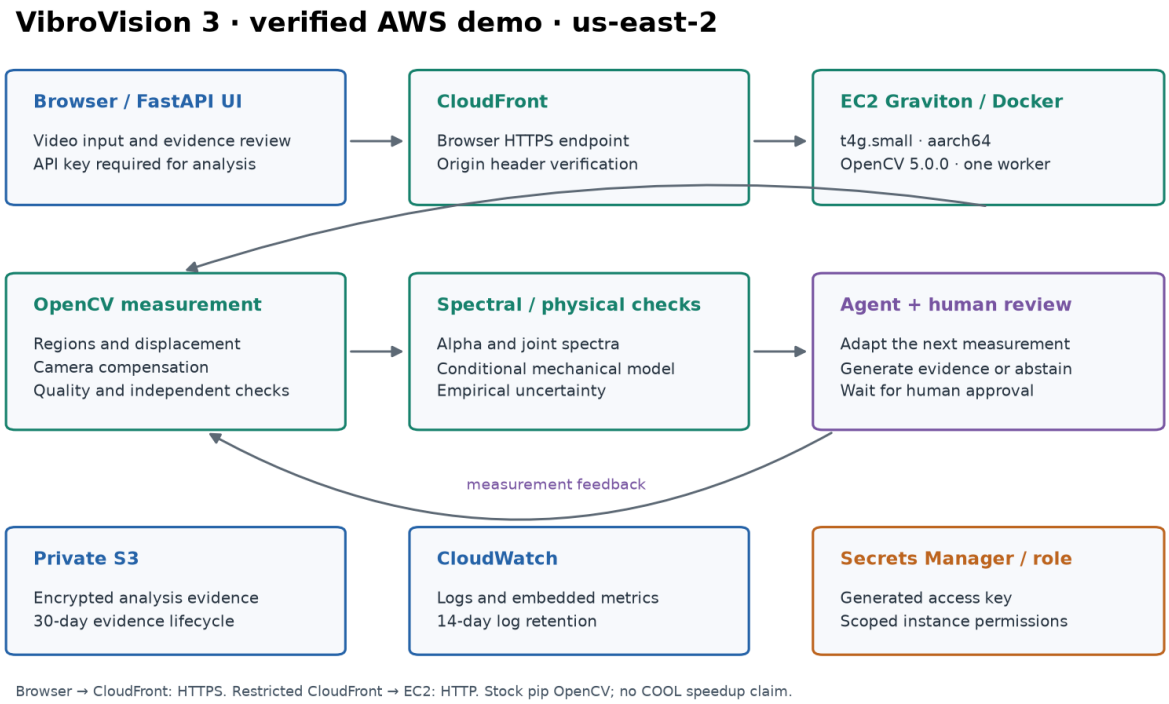


Original bundled fan recordings: without a coin (left), with a coin (right). Source hashes are attached to the recorded AWS result.

Evidence	Recorded result	Interpretation
Real fan comparison	35.9x amplitude ratio; 43.9 s wall time	Large imposed perturbation in these two recordings
Synthetic video recovery	2.03% relative operator error	One controlled, model-compatible benchmark
Automated checks	60 passed	Math, tracking, API, video lab and packaging

Central limitation: the approximately 15-second fan clips observe one housing. Their alpha ranges overlap. They demonstrate a motion change and **abstain from identifying a damaged part or reconstructing full machine stiffness**. The amplitude ratio is not a field sensitivity or speedup benchmark.

Vision and AWS architecture



Deployed application and measurement feedback loop. Browser access uses HTTPS; the restricted CloudFront-to-EC2 origin connection uses HTTP.

Substantive image and video analysis

OpenCV 5 tracks image features at subpixel precision, compensates camera motion against a static reference, checks ROI quality and independently cross-validates with ECC. In paired fan clips, detected housing motion is measured relative to the background before comparing spectra and tone-band displacement.

Compute: Python 3.13 / Docker on EC2 t4g.small / aarch64 in us-east-2. OpenCV 5.0.0 executes the image-analysis workload.

Evidence and access: private encrypted S3 evidence expires after 30 days. CloudWatch retains logs for 14 days. Secrets Manager stores the application key; an instance role provides scoped permissions.

One worker processes jobs sequentially with a 512 MiB frame-buffer limit. Raw uploads are deleted after processing; retained evidence follows its lifecycle. In-memory web history resets after a restart. Confidential production use needs origin TLS and user isolation.

Deployment source, pinned dependencies and lifecycle instructions: infra/ and DEPLOYMENT_AWS.md.

What the mathematics adds

The mathematics distinguishes a visible change from a supported structural interpretation. Signal spectra, mechanical modes, damping and graph connectivity answer different questions; no single spectral slope or connectivity score is a universal damage indicator.

Frequency content and relationships between regions

Welch power spectra summarize how motion is distributed over frequency. The video sampling rate limits resolvable frequencies. A joint spectral matrix also retains relationships and relative phase between channels: equally strong motions may be coordinated or opposed. Its operational patterns are not automatically natural modes or proof of damage.

A conditional, scale-invariant mechanical operator

$$M\,x'' + C\,x' + K\,x = f \qquad y = Sx + \eta$$

Assume reciprocal linear spring couplings, positive diagonal mass M, fixed positive diagonal image scales S, sufficient observed coordinates and identifiable modes. The pixel-coordinate stiffness operator is:

$$B_y = S\,M^{-1}\,K\,S^{-1}$$

$$D_{ii} = (B_y)_{ii} \qquad D_{ij} = -\sqrt{[(B_y)_{ij}(B_y)_{ji}]},\ i \neq j$$

$$D = M^{-1/2}\,K\,M^{-1/2}$$

The reciprocal product cancels the fixed positive region scales. In the illustrative unit-mass model, changing the two coefficients from -2/-2 to -4/-1 leaves the product 4 and recovered coupling -2 unchanged. Diagonal entries are retained. This does not cancel arbitrary camera transformations or recover calibrated physical masses, forces or stiffness separately.

Natural modes and uncertainty

$$D\,\varphi_k = \omega_k^{\,2}\,\varphi_k$$

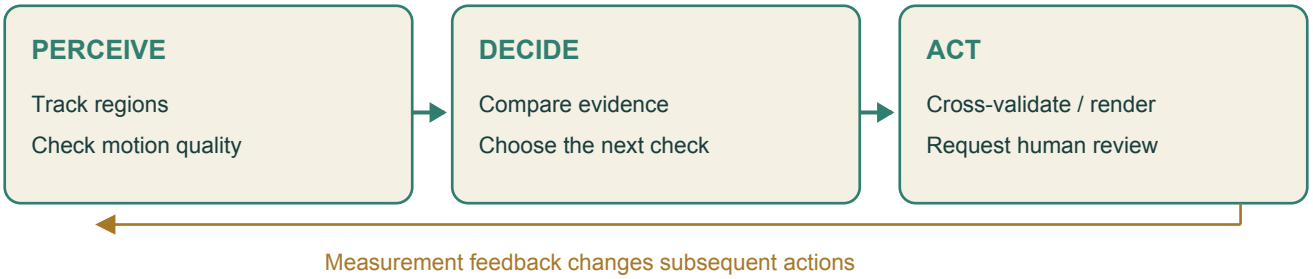
Eigenvalues describe squared natural angular frequencies of the compatible conservative mechanical model; eigenvectors describe mass-normalized mode shapes. Physical signs, positivity, conditioning and compatibility gate operator acceptance.

Empirical multichannel block-bootstrap radii inform whether a change or localization is supported. They are conditional estimates without certified coverage and exclude model bias and nonstationarity. Unknown forcing can prevent causal structural identification; one spectral peak is insufficient to reconstruct the operator.

Derivations and implementation conditions: full mathematical appendix (Spanish). Exact analytic video examples: video mathematics guide (English).

Agentic workflow and evidence

Requested special-award consideration: **Agentic Vision Award**. The workflow uses OpenCV outputs to alter later tool calls and request human review. It is rule-based; no LLM is claimed as the inspection controller.



Recorded AWS trace: controlled synthetic fault, seed 7

Video time	Tool	Observed next step
88 s	watch	Alpha drift triggers closer monitoring.
110 s	cross_validate	Independent measurement agrees; suspicion is escalated.
133 s	render_evidence	Generate spectrum and magnified x-t evidence.
133 s	request_human_approval	Create a review request; wait for a person.

Job 7b61959df6ab; 80.6 s observed wall time. Trace times above are analyzed-video time, not processing latency. The captured run contains 20 trace events and 1 incident awaiting review. This is synthetic evidence, not field sensitivity.

Failure handling, observability and human control

The implementation can reassess a region, extend a window and cross-check an estimate. Invalid uploads and excessive frame-buffer requests return errors. Insufficient or incompatible structural observations can lead to more data or abstention. The UI exposes measurements, tool decisions and evidence; logs and metrics support inspection of the hosted execution.

A work-order recommendation needs an explicit reviewer decision. In a new synthetic job, a judge can enter a reviewer name and select Reject to verify that the incident becomes REJECTED. This review test is documented separately from the captured pending incident. No machinery is controlled.

Award scope: the submitted runtime uses stock pip OpenCV. COOL integration, Graviton performance speedup and a cost benchmark are not claimed. The Best Use of COOL Award is not requested.

Award requirements: official competition requirements. Supporting workflow source and trace: vibrovision/agent.py; results/aws_agent_trace.json.

Testing and reproducibility

Fast judge walkthrough

1. Access: open the live endpoint on page 1. Paste the separately supplied judge API key into the API key field, click outside it and wait for Ready. No AWS account is needed. This report intentionally contains no credentials.

2. Real recordings: click Compare the two real fan videos and wait for done. Inspect the two cards and figure. The recorded AWS reference ratio is 35.9x; the structural verdict is abstention. This button runs fresh analysis.

3. Agent and human gate: leave the upload empty, select fault, keep seed 7 and click Start inspection. Inspect the trace and incident evidence. In your new test, enter JudgeTest and click Reject; confirm REJECTED. Allow a few minutes per run; the server has one worker.

Run tests and regenerate mathematical evidence locally

```
git clone https://github.com/emezac/vibrovision.git
cd vibrovision
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-dev.txt
python -m pytest -q
python eval/run_math_validation.py --real \
  --out results/judge_math_validation.json
uvicorn service.app:app --host 127.0.0.1 --port 8000
```

Use Python 3.12+ and FFmpeg. The reference suite has 60 passing tests. Validation should report algebra: true and synthetic_video: true and write JSON plus a figure. The local app opens at http://127.0.0.1:8000 and needs no API key by default unless API_KEY is set. Windows activation: .venv\Scripts\activate.

Evidence and further reading

Full judge guide, local fallback and optional Drive/Colab setup

AWS comparison, source hashes and measured values

Synthetic agent trace and pending human review

Algebra, 100 scale trials and synthetic-video validation

Evidence index and historical execution boundaries

4:33 demonstration video and English captions

Remaining work: validate longer recordings of distinct parts against independent sensors; vary loads, lighting, compression and camera conditions; calibrate uncertainty and false alarms; evaluate detection, localization and abstention separately. The MIT/LISS montage is not an independent healthy/damaged pair. See the technical report and evidence index for details.

Public evidence baseline: repository revision e880376. The earlier local fan result (~36.7x) is a separate execution from the recorded AWS result (35.9x). Recorded values and test counts are reference evidence, not new executions of every experiment.